Temporal Prototype Alignment

for Frozen-Feature Dataset Distillation

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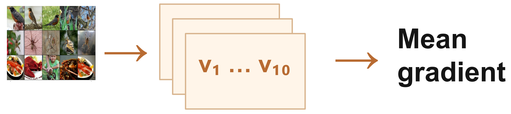
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Project page

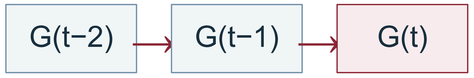
1 Why temporal reuse?

LGM stabilizes classifier gradients by averaging ten augmented views per step through frozen encoders. Processing these views together increases memory and I/O costs.

LGM: evidence across space



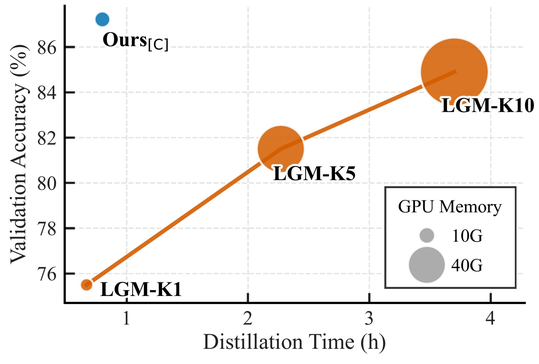
TPA: evidence across time



One fresh view + a running prototype

Fixed-size moments reuse gradient history across steps instead of concurrent views.

Memory and accuracy

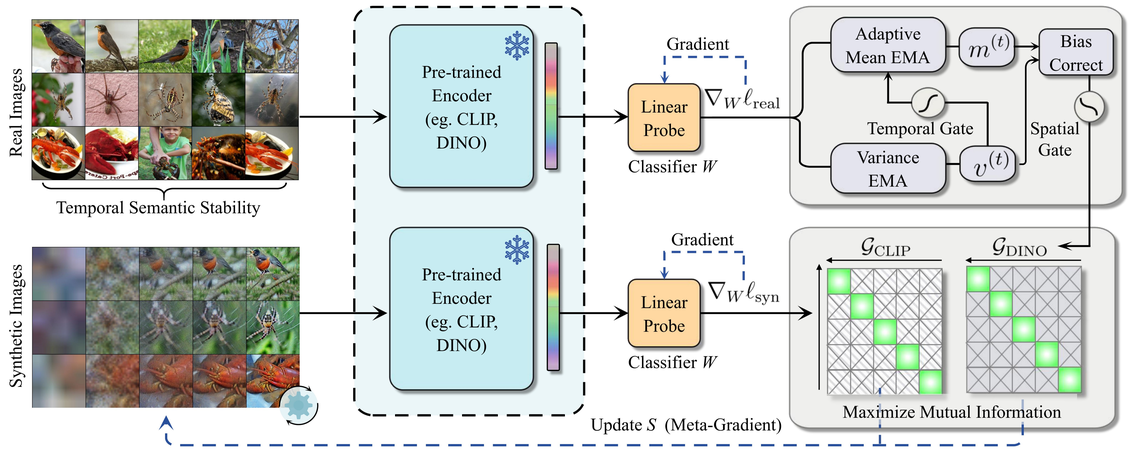


81 → 7 GB **3.6 → 0.8 h** **84.9 → 86.7%**

CLIP, ImageNet-100, IPC = 1.
LGM: 4 × RTX 4090. TPA: 1 × RTX 4090.

2 How temporal prototypes guide synthesis

Real-data gradients build temporal prototypes. Frozen teachers align synthetic gradients to these targets. Only synthetic pixels are updated.



Upper path: temporal prototypes. Lower path: teacher-local class alignment.

Adaptive Temporal Momentum (ATM)

ATM accumulates first and second moments from one view per step. A **reliability-dependent gate** controls how new gradients update the history. **Inverse-RMS weighting** builds the prototype. Bias correction accounts for zero initialization. Real and synthetic batches use the same freshly sampled classifier head at each step.

Contrastive Consensus Alignment (CCA)

CCA aligns class-wise synthetic gradients with real prototypes **within each teacher**. Same-class pairs are positives. Other classes are negatives. CLIP, DINOv2 and MoCo-v3 jointly optimize **one shared image set**. Each teacher keeps its own head and prototypes. Alignment stays within each feature space, allowing teachers with different dimensions.

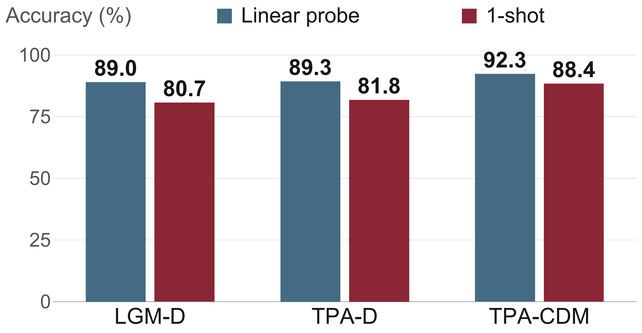
3 Cross-encoder transfer

ImageNet-100, one image per class (IPC = 1). One distilled set is evaluated on nine unseen encoders with linear probing and training-free 1-shot inference.

+3.3 pp Linear probing

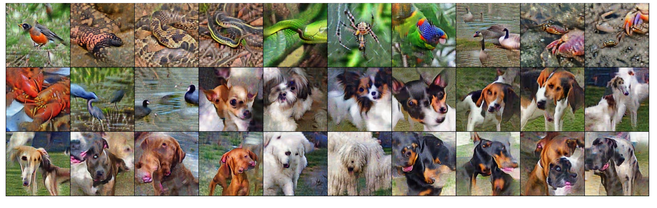
+7.7 pp 1-shot inference

Gain over LGM-D · mean of nine unseen encoders



TPA-CDM reaches **92.3%** linear-probe and **88.4%** 1-shot accuracy across nine unseen encoders. Multi-teacher synthesis improves both transfer metrics over LGM-D.

What does the distilled set look like?



Selected images from the paper's TPA-CDM set.